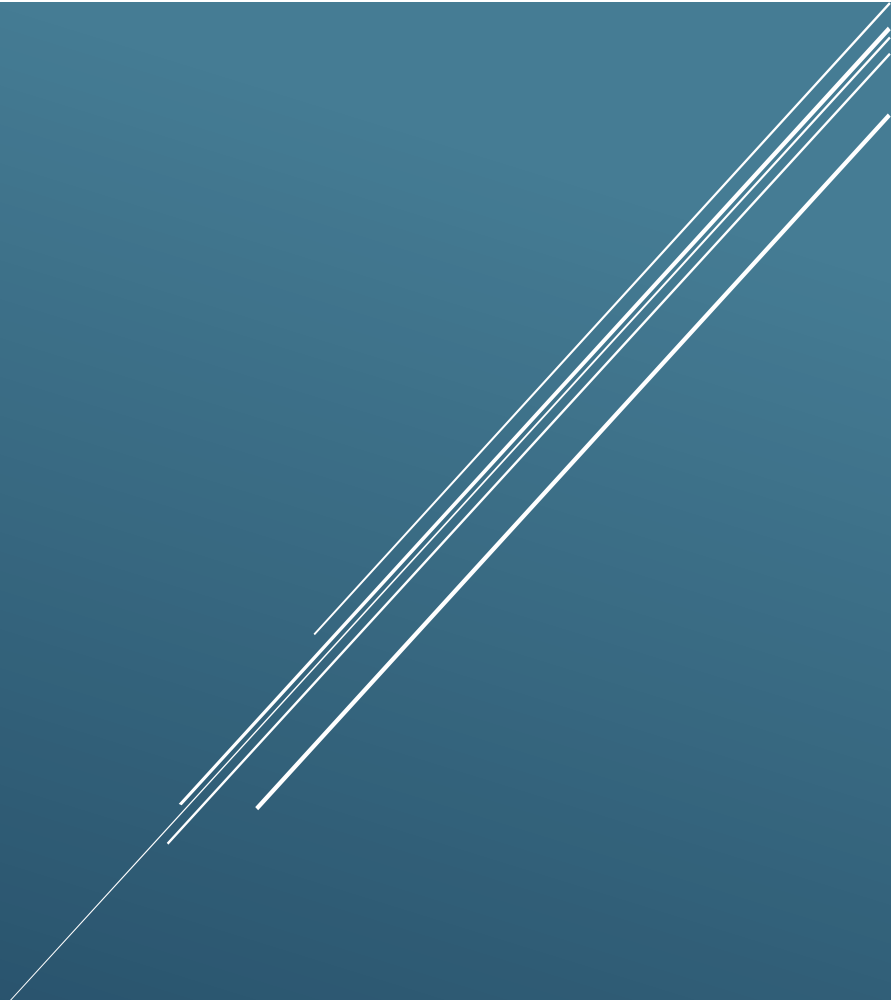




CUT DIGITAL MAINTENANCE COMPLAINT SYSTEM

Phase 1: Planning, Requirements and Feasibility —
ITC327W Group Assessment 1

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1.1 Stakeholder and Organisation

The organisation addressed in this project is the Central University of Technology (CUT), with particular focus on the Estate and Infrastructure Department and the maintenance of CUT residences and other institutional units. The primary stakeholder is CUT Estate and Infrastructure because this unit is responsible for receiving and processing maintenance complaints and generating job cards for maintenance work. The project specifically responds to the gap between the reporting of a maintenance problem and the successful confirmation that the problem has been resolved within an expected timeframe.

The project therefore treats maintenance complaint management as an end-to-end organisational process rather than merely a method of submitting a complaint. The key users identified are House Committee Members, Estate and Infrastructure supervisors, and maintenance employees. System administrators will also be involved in managing authorised accounts, roles and configurable maintenance categories.

1.2 Stakeholders and People Involved

- House Committee Members (HCMs): The reporters. They identify maintenance problems, submit complaints, provide information about the location and nature of the problem, attach photographic evidence where appropriate, track progress and verify whether the reported problem has been resolved.
- Estate and Infrastructure Supervisors: receive incoming complaints, categorise and prioritise them, assign work to maintenance employees, set response or attendance deadlines, monitor progress and escalate overdue complaints.
- Maintenance Employees: receive assigned maintenance jobs, accept the work, update progress and mark work as completed once the maintenance activity has been performed.
- System Administrators: authorised system administrators manage user accounts, roles and configurable maintenance categories.

1.3 Current Maintenance Complaint Process

The current process is primarily manual and relies on email for reporting maintenance issues. The Estate and Infrastructure Unit then processes the complaint and generates a job card. Although this enables maintenance work to be initiated, the process does not provide a complete digital lifecycle from the initial report through to verification and closure.

Stage	People Involved	Limitation Identified
Complaint reporting	House Committee Member; Estate and Infrastructure	Email reporting causes information to be fragmented.

Stage	People Involved	Limitation Identified
Job-card generation	Estate and Infrastructure supervisors	Initiates the maintenance work but does not provide a complete digital complaint lifecycle.
Follow-up	House Committee Member; supervisors	Manual follow-up provides limited deadline monitoring and escalation.
Maintenance attendance	Maintenance employees; supervisors	The process does not comprehensively integrate assignment, status monitoring and deadline control.
Completion	Maintenance employee; complainant	A maintenance employee marking a job complete does not by itself confirm successful resolution from the complainant's perspective.
Historical records	Estate and Infrastructure management	Scattered historical information makes recurring problems and performance difficult to analyse.

The proposed improvement is an integrated lifecycle: Report → Prioritise → Assign → Set deadline → Monitor → Escalate → Attend → Complete → Verify → Close/Reopen → Store history → Analyse. This demonstrates that the intended system is not simply an electronic complaint form, but a system for managing the entire maintenance complaint lifecycle.

1.4 Problem Experienced

The central problem experienced by CUT is inefficient and fragmented maintenance complaint management resulting from the absence of a centralised digital system for reporting, assigning, monitoring, escalating, resolving and evaluating maintenance complaints. The current manual process can initiate maintenance work, but it does not comprehensively connect complaint reporting with prioritisation, assignment, response-time monitoring, escalation, resolution tracking, complainant verification and historical analysis.

This creates a gap in accountability and visibility. A complaint may be reported and converted into a job card, but the available process does not provide a single digital view showing the complaint's complete progress, whether the response deadline has been exceeded, whether the work was resolved, or whether the complainant is satisfied with the outcome.

1.5 Consequences of the Problem

- Fragmented reporting and limited visibility of the progress of individual maintenance complaints.
- Reduced accountability for response and attendance deadlines because monitoring and follow-up are limited.
- Difficulty identifying overdue complaints and escalating them to the appropriate supervisor.
- A completed job may not represent a successfully resolved problem because complainant verification is limited absent.
- Difficulty using historical maintenance information to identify recurring problems.
- Reduced ability for management to compare response and resolution times and use evidence when making future maintenance and resource allocation decisions.

1.6 Affected Users

The problem affects multiple groups because the existing process involves several parties and does not provide one integrated source of information.

- House Committee Members are affected because they have limited visibility of complaint progress and may need to rely on manual follow-up.
- Estate and Infrastructure supervisors are affected because complaints, deadlines, assignments and escalation activities are not comprehensively integrated in one digital workflow.
- Maintenance employees are affected because job assignment and progress information are not managed through a complete lifecycle-management process.
- CUT management is affected because historical complaint information is not effectively consolidated for identifying recurring faults, evaluating performance and supporting informed decisions.

1.7 Evidence that the Problem and Stakeholder Needs Were Investigated

The problem and needs described above were investigated through engagement with the Estate and Infrastructure environment and through analysis of the existing email-based reporting and job card process. Following are the meeting notes with the Estate and Infrastructure manager Mr. Khumalo

Meeting Notes:

Meeting with Estate and Infrastructure Unit CUT

Date: 19 August 2026

Venue: Estate and Infrastructure main gate offices

Stakeholder: Mr Khumalo – Estate and Infrastructure Unit

Purpose: To understand the department's current system, challenges, and requirements for the proposed project.

Discussion

We had a meeting with Mr Khumalo from the Estate and Infrastructure Unit to understand how the department currently operates and what they need from the proposed system.

Mr Khumalo explained how their current system works and highlighted some of the challenges they experience with the existing process. He also explained the type of information that needs to be captured and managed by the department.

Requirements Discussed

Mr Khumalo explained that the new system should:

- Keep proper records of reported issues.
- Allow users to provide information about the problem, including the location and description.
- Make it easier for the Estate and Infrastructure Unit to track reported issues.
- Allow staff to update the status of a request, whether it is pending, in progress, or completed.
- Make it easier to see which issues still need attention.
- Keep information organised so that it can be accessed when needed.
- Improve communication between the people reporting problems and the Estate and Infrastructure Unit.
- Provide a simple and user-friendly interface.
- Protect the information stored in the system and ensure that only authorised people can access certain information.

Current System

Mr Khumalo also explained the process that the Estate and Infrastructure Unit is currently using, and it is called Maintenance Management Systems (MMS). We discussed how requests are currently received, recorded, assigned, and followed up.

The current system works as follows, the reporter must report to the help desk via an email or phone call. The supervisor will then generate a job card from their current system, assign the job card to the maintenance workers, then the execution happens.

The meeting helped us understand that the proposed system should not simply replace the current process, but should improve it by making the reporting, tracking, updating, and management of issues easier and more organised.

Conclusion

The meeting was useful because Mr Khumalo gave us a better understanding of the Estate and Infrastructure Unit's current system, their challenges, and the requirements for the proposed system.

1.8 Stakeholder Needs Identified

Stakeholder Need	Required System Response
Easy reporting	A simple Flutter complaint form for authorised House Committee Members.
Issue evidence	Ability to take or select a photograph and associate it with the complaint.
Classification	Maintenance categories
Visibility	An administrator dashboard and notifications for new and relevant workflow events.
Accountability	Assignment, deadlines, statuses and escalation of overdue work.
Resolution confirmation	Complainant verification after maintenance staff mark work as completed.
Historical information	A central complaint and status/action history.
Decision support	Basic analysis by category, residence and time, including response/resolution measures where data exists.

1.9 Clear Problem Statement

Central University of Technology (CUT) currently manages maintenance complaints from residences through an email-based process that is used by Estate and Infrastructure to initiate job-card generation. However, this process does not provide a comprehensive, integrated mechanism for managing the complete maintenance complaint lifecycle, including reporting, categorisation, prioritisation, assignment, response-time monitoring, escalation, resolution tracking, complainant verification and historical analysis. Consequently, CUT experiences fragmented information, limited visibility and accountability, difficulty identifying overdue work, limited confirmation of successful resolution, and difficulty using historical maintenance data for recurring-problem identification, performance evaluation and informed planning. Therefore, there is a need for a centralised digital maintenance complaint lifecycle management system that enables complaints to be reported, assigned, monitored, escalated, resolved and verified while retaining a reliable historical record for future analysis and decision-making.

1.10 Summary of the Analysis

The stakeholder and problem analysis establishes that the issue at CUT is not simply the absence of an online complaint form. The fundamental issue is the lack of an integrated end-to-end process for managing maintenance complaints after they are reported. The principal stakeholder environment is CUT Estate and Infrastructure, while House Committee Members, supervisors, maintenance employees and system administrators interact with the process in different capacities. This analysis forms the foundation for the requirements, scope and feasibility decisions developed in the remainder of this document.

2 Software Requirements Specification

2.1 Project Introduction

The Digital Maintenance Complaint Lifecycle Management System is a proposed integrated solution for Central University of Technology (CUT), with particular focus on residences. The project is intended to improve the way maintenance complaints are reported, processed, monitored, resolved and verified. The Estate and Infrastructure Department is the principal organisational stakeholder because it receives and processes maintenance complaints and generates job cards.

The project responds to a gap identified in the existing process: although maintenance complaints can be submitted manually, primarily through email, and can be processed for job-card generation, the current process does not provide a complete digital lifecycle from the initial report to successful resolution and closure. The proposed solution therefore moves beyond a simple online complaint form and introduces an integrated lifecycle covering reporting, prioritisation, assignment, deadlines, monitoring, escalation, attendance, completion, complainant verification, closure or reopening and historical analysis.

The proposed technical solution consists of a Flutter mobile application using MVVM for authorised House Committee Members, an ASP.NET Core web application for Estate and Infrastructure administration, and a shared Supabase backend. The two applications are intended to operate as one integrated solution through the shared backend.

2.2 Stakeholder Profile and Current Process

The organisation is Central University of Technology (CUT), particularly the Estate and Infrastructure Department and the maintenance environment serving residences and other institutional units. The key users are House Committee Members, Estate and Infrastructure supervisors, maintenance employees and system administrators.

The current process is primarily email complaint reporting followed by Estate and Infrastructure processing and job-card generation. Further follow-up is largely manual. Completion is not necessarily followed by formal complainant verification, while historical information is scattered and difficult to use for recurring-problem identification and

performance evaluation. The key limitation is that the existing process initiates maintenance work but does not comprehensively integrate reporting, categorisation, prioritisation, assignment, deadline monitoring, escalation, resolution tracking, verification and historical decision-making.

2.3 Problem Statement

CUT currently manages maintenance complaints from residences and other institutional units through a process that is primarily manual and email-based before Estate and Infrastructure processes the complaint for job-card generation. While this facilitates the initiation of maintenance work, it does not provide a comprehensive digital mechanism for managing the complete lifecycle of a maintenance complaint.

The absence of end-to-end integration creates fragmented reporting, limited visibility of complaint progress, reduced accountability for response and attendance deadlines, difficulty identifying and escalating overdue work, and limited confirmation that a completed job has resolved the complainant's problem. Historical maintenance information is also not effectively consolidated into a central dataset for identifying recurring problems, evaluating maintenance performance and supporting informed planning.

The affected users include House Committee Members, Estate and Infrastructure supervisors, maintenance employees and CUT management. CUT therefore needs a centralised digital maintenance complaint lifecycle management system capable of reporting, assignment, monitoring, escalation, resolution, verification, closure or reopening and historical analysis.

2.4 Project Aim and Objectives

Aim: To develop an integrated digital maintenance complaint lifecycle management system for CUT residences and institutional units that centralises reporting, categorisation, assignment, deadline monitoring, escalation, resolution, verification, closure or reopening and historical maintenance records.

Objectives:

- To design and build a Flutter mobile application through which authorised House Committee Members can report maintenance complaints with supporting details and photographic evidence.
- To design and build an ASP.NET Core web application through which Estate and Infrastructure staff can categorise, prioritise, assign, monitor and escalate complaints.
- To implement a shared Supabase backend that keeps complaint, assignment, status and history data consistent between the two applications.

- To implement complainant verification so that a completed job can be confirmed, partially confirmed or reopened.
- To retain historical complaint records and provide basic search, filtering and analytics to support management decisions.

2.5 Project Scope

Deliverables

The project will deliver an integrated prototype consisting of a Flutter MVVM mobile application for House Committee Members, an ASP.NET Core web application for Estate and Infrastructure administration, and a shared Supabase backend. The system will support authentication and role-based access, complaint submission, category/location/photo capture, assignment, deadlines, monitoring, overdue escalation, notifications, completion, complainant verification, closure/reopening, historical records, administrative search and basic analytics.

Acceptance criteria:

- An authorised House Committee Member can log in and submit a complaint with required details and a photograph.
- An administrator can view and classify a complaint.
- An administrator can assign the complaint and set a deadline.
- The defined lifecycle (Reported → Assigned → Accepted → In Progress → Completed → Verified → Closed) can be demonstrated end-to-end.
- Overdue complaints can be identified and escalated.
- Maintenance employees can mark work completed.
- Complainants can verify, partially verify or reject the resolution.
- Unresolved complaints can be reopened.
- Closed complaints remain in history.
- Administrators can search by category, status, residence and date.
- Flutter and ASP.NET can operate against the shared Supabase backend as one integrated solution.

Exclusions

- Full enterprise asset-management functionality not required by the case.
- Procurement, invoicing and payment processing.
- Automatic fault diagnosis from photographs.
- External contractor integration, unless confirmed by the stakeholder.
- Advanced predictive maintenance.

Constraints and Assumption

- The semester timeframe requires disciplined scope control.
- Users need network access to synchronise with the backend.
- Protected complaint information must be accessible only to authorised roles.
- The team must manage integration risk across Flutter, ASP.NET and Supabase

2.6 Stakeholder Needs

Easy reporting will be addressed through a simple Flutter complaint form. Issue evidence will be supported through camera photo capture. Classification will use configurable maintenance categories. Visibility will be supported through an administrator dashboard and notifications. Accountability will be strengthened through assignment, deadlines, statuses and escalation. Resolution confirmation will be supported by complainant verification. Historical information will be retained through central complaint and status history. Decision support will be supported through analysis by category, residence and time.

2.7 Functional Requirements

ID	Function	Requirement
FR-01	Authentication	Authenticate authorised users.
FR-02	Roles	Restrict functions according to the user's role.
FR-03	Report	Allow an authorised House Committee Member to create a complaint.
FR-04	Details	Capture title, description, category, location, reporter and date/time.
FR-05	Photo	Allow a photo to be taken/selected and associated with a complaint.
FR-06	Categories	Support maintenance categories;
FR-07	ID	Generate a unique complaint ID.
FR-08	Notifications	Notify relevant administrators of new complaints and relevant workflow events.
FR-09	Prioritise	Allow an authorised supervisor to set priority.
FR-10	Assign	Assign a complaint to a maintenance employee.
FR-11	Deadline	Set response deadlines.

ID	Function	Requirement
FR-12	Lifecycle	Support Reported, Assigned, Accepted, In Progress, Completed, Verified and Closed statuses.
FR-13	Overdue	Identify complaints exceeding their deadline.
FR-14	Escalation	Escalate overdue complaints to the responsible supervisor.
FR-15	Complete	Allow maintenance employees to mark work completed.
FR-16	Verify	Allow the complainant to select Resolved Successfully, Partially Resolved or Not Resolved.
FR-17	Reopen	Allow Not Resolved complaints to be reopened.
FR-18	History	Record important status/action changes with timestamps and actor.
FR-19	Historical	Retain closed complaints as historical records.
FR-20	Filter	Filter by category, status, residence and date.
FR-21	Search	Search by complaint ID and relevant text.
FR-22	Dashboard	Show total, new, active, overdue and closed summaries.
FR-23	Analytics	Analyse complaints by category/residence/time and response/resolution metrics where available.
FR-24	Categories Admin	Allow authorised administrators to maintain categories.

2.8 Non-Functional Requirements

ID	Area	Requirement
NFR-01	Usability	Complaint submission should be simple and minimise unnecessary steps.
NFR-02	Performance	Normal screens/lists should respond within practical prototype conditions.
NFR-03	Security	Authentication and role-based access must protect administrative functions and data.
NFR-04	Availability	The system should be available when users have network.

ID	Area	Requirement
NFR-05	Integrity	Complaint, assignment, status and history relationships must remain consistent.
NFR-06	Maintainability	Use Flutter MVVM and an organised ASP.NET application structure.
NFR-07	Scalability	Support additional residences, categories, users and complaints.
NFR-08	Privacy	Collect and expose only information necessary for maintenance management.
NFR-09	Auditability	Important workflow changes must be traceable.
NFR-10	Portability	Flutter should support the intended mobile environment; ASP.NET should run in a supported browser.

2.9 Integration Requirements

The assessment requires Flutter, ASP.NET and Supabase to operate as one integrated solution through a shared Supabase backend. Flutter/MVVM will provide the House Committee Member-facing mobile functions, including login, complaint reporting, camera or gallery access, complaint tracking, notifications and verification. ASP.NET Core will provide the Estate and Infrastructure administration functions, including administrator login, dashboard, complaint management, categorisation, assignment, deadlines, monitoring, history and analytics. Supabase will provide the shared authentication layer and storage for complaint photographs, subject to final implementation decisions.

The core information model proposed includes Users, Complaints, Categories, Complaint Images, Assignments, Complaint History and Notifications. The central lifecycle is Reported → Assigned → Accepted → In Progress → Completed → Verified → Closed. If the complainant selects Not Resolved, the complaint can be reopened for further action.

Integration must preserve a shared source of complaint information so that an action taken in one application is reflected in the other application through the backend. Authentication and role-based access must restrict functions according to the user's role, while complaint history must retain important workflow changes with timestamps and the responsible actor.

2.10 Requirements Change Log

The requirements were refined after considering the case framing, the feasibility study and the project risk register. The changes below make the proposed system more realistic and aligned with the end-to-end problem, rather than treating the project as a simple complaint form.

- All CUT institutional units → CUT residences only: the system was initially planned to allow House Committee Members and other stakeholders across all CUT institutional units to lodge maintenance complaints. The schedule feasibility analysis identified this as a significant risk to on-time delivery, because it would require the group to investigate, confirm and design stakeholder needs, categories and reporting workflows for multiple additional units within a fixed semester timeframe. The scope was therefore narrowed to CUT residences only the primary environment already investigated in the stakeholder and problem analysis so that the group could deliver a complete, working lifecycle for one well-understood stakeholder group rather than a partial system spread across many units. Extending the system to other CUT institutional units remains a realistic future enhancement beyond Phase 1.
- Added complainant verification: staff completion alone does not prove that the problem was successfully resolved, so verification and reopening were added.
- Configurable categories: categories require stakeholder confirmation and may change, so the category list is not hard coded as a final business rule.

2.11 Refined Project Scope

The realistic scope taken forward into system design is an integrated prototype comprising:

- (1) a Flutter MVVM mobile application for House Committee Members
- (2) an ASP.NET Core web application for Estate and Infrastructure supervisor
- (3) a shared Supabase backend.

The baseline will implement complaint reporting with category, location, description and photo capture; assignment and priority management; response, status tracking, overdue identification and escalation; completion by maintenance employees; complainant verification; closure and reopening; historical storage, notifications; administrative search, and basic analytics where achievable within the semester schedule.

Advanced enterprise asset management, automatic image-based diagnosis, external contractor integration and advanced predictive/AI features will remain excluded. This refined scope is intended to protect the core lifecycle from the project's identified schedule and integration risks while still addressing the central stakeholder problem.

3. FEASIBILITY STUDY

The feasibility study evaluates whether the proposed stakeholder project can realistically be developed and implemented within the available resources, skills, time, and constraints of the group. We assessed the project across six key feasibility areas: technical, operational

, economic, schedule, legal, and ethical feasibility.

Technical Feasibility

The proposed project is technically feasible because we have access to the necessary technology, equipment, software, and development tools required to design and produce the prototype. We can make use of standard computers, internet connectivity, development software, and database technology, which is Supabase in this case, to develop and demonstrate the proposed solution.

Limitations may stem from our lack of technical experience and skills. To overcome this, we will allocate tasks according to one's individual strengths

Operational Feasibility

The project is workable because it solves the stakeholder's real problems and makes processes easier and information more accessible. The system will be simple to use, with a clear interface that doesn't require technical skills.

The main challenge is that users might resist change or be unfamiliar with the new system at first. This can be managed with a user-friendly design, basic guidance, demonstrations, and feedback during development.

The prototype should reflect how stakeholders work, not just show technical features.

Economic Feasibility

The project is affordable because it doesn't need much money. We will use existing computers, internet, and free or provided software.

Main resources are computers, internet, software, design tools, databases, and group members' time. Costs like hosting, printing, or special software are not necessarily needed

The biggest risk is poor time management.

Schedule Feasibility

The project can fit into the semester because there is enough time to research, design, build, test, and present the prototype.

The main challenge is the fixed academic deadline; delays in gathering requirements, communicating with stakeholders, and developing and testing could cause problems.

To stay on track, we will break the work into smaller tasks and set earlier internal deadlines to allow time for corrections and feedback.

Ethical Feasibility

The project is ethically acceptable as it will protect stakeholder and user rights. The system will treat all users fairly, avoid misuse of stakeholder information, and will be transparent about what data is collected and why.

It will be accessible so that as many intended users as possible can use it without barriers.

Overall Feasibility Decision

We as the group believe this project can be done with the skills and tools we already have. It will help the Estate and Infrastructure Unit with the problems they identified.

The prototype will not cost much to build, and we can finish it within the semester if we manage our time well, share tasks, and check progress often.

We must pay close attention to the technical setup, time management, data privacy, user access, and safe handling of information. We will also keep in touch with the stakeholder and use their feedback to make sure the system meets their needs.

4. Project Risk Register

4.1 Risk Summary

ID	Risk Statement	Category	Likelihood	Impact	Score	Owner	Status
R-01	Insufficient time to deliver the full complaint lifecycle within the semester.	Schedule	High	High	High	Project Lead	Open
R-02	Flutter–ASP.NET–Supabase integration delay.	Technical	Medium	High	High	Technical Lead	Open
R-03	Incorrect deadline logic produces wrong overdue status.	Technical	Medium	Medium	Medium	Backend Developer	Open
R-04	Limited stakeholder availability for confirming deadlines and rules.	Stakeholder	Medium	High	High	Stakeholder Liaison	Open
R-05	Uneven group skill levels across Flutter, ASP.NET and Supabase.	Skills	Medium	Medium	Medium	Team Lead	Open
R-06	Loss or corruption of complaint photo evidence in storage.	Technical	Low	Medium	Low-Med	Backend Developer	Open

ID	Risk Statement	Category	Likelihood	Impact	Score	Owner	Status
R-07	Unauthorised access to complaint or personal data.	Security/Legal	Low	High	Medium-High	Technical Lead	Open
R-08	Scope creep from additional stakeholder requests beyond the agreed baseline.	Scope	Medium	Medium	Medium	Project Lead	Open
R-09	Version-control conflicts or lost work in the GitHub repository.	Project Management	Medium	Low	Low-Med	Repository Manager	Open
R-10	Network/connectivity issues affecting Supabase sync during testing or demonstration.	Operational	Medium	Medium	Medium	Tester	Open
R-11	Miscommunication of requirements leading to rework.	Communication	Medium	Medium	Medium	Business Analyst	Open
R-12	Uneven individual contribution among group members.	Team/People	Medium	Medium	Medium	Team Lead	Open

4.2 Mitigation and Contingency Actions

ID	Mitigation Action	Contingency Action
R-01	Prioritise the core lifecycle first; build in small, demonstrable increments; set internal milestones ahead of the assessment deadline.	Reduce scope to core reporting, assignment and completion only; defer analytics and advanced filtering.
R-02	Complete an early end-to-end integration spike (one screen through Flutter → ASP.NET → Supabase) before building further features; define shared API/data contracts up front.	Fall back to direct Supabase client calls from both applications if the planned integration approach stalls.

ID	Mitigation Action	Contingency Action
R-03	Unit test deadline and escalation boundary conditions; review logic with more than one group member.	Allow an administrator to manually override status/escalation if automated logic is incorrect.
R-04	Schedule short, recurring stakeholder check-ins; use email/Teams for asynchronous confirmation of decisions.	Proceed with clearly documented assumptions (Section 1.11) pending confirmation, and update the requirements once confirmed.
R-05	Pair less-experienced members with more experienced ones; allocate tasks according to individual strengths; share short internal tutorials.	Reassign a struggling task to another member; have a senior member review or complete critical integration work.
R-06	Use Supabase storage with validated upload types/sizes; avoid relying on local-only storage.	Allow re-upload of the photo and retain complaint text/metadata independently of the image.
R-07	Enforce authentication and role-based access control on all administrative functions, apply least-privilege access.	Maintain audit/history logs to detect misuse; revoke and reset any compromised account.
R-08	Maintain the Requirements Change Log; require sign-off before adding new requirements to the baseline.	Log new stakeholder requests in a backlog for consideration after Phase 1 rather than expanding the current baseline.
R-09	Use a simple branching strategy, commit regularly with clear messages, and review changes before merging.	Restore from repository history or a backup branch if work is lost or overwritten.
R-10	Test the applications on both campus and off-campus networks ahead of the demonstration; design the UI to handle slow or failed sync gracefully.	Prepare a local demonstration or short recorded walkthrough as a fallback for the live presentation.
R-11	Confirm requirements in writing after each stakeholder interaction; review requirements as a group before implementation.	Re-baseline the affected requirement with the stakeholder and update the SRS and change log accordingly.
R-12	Track task allocation and completion in the ITC327W Project Progress Tracker; hold brief weekly check-ins.	Escalate persistent non-contribution to the module coordinator in line with university procedure.

Stakeholder evidence

1. Interview questions:

- Can you walk me through what happens, step by step, from the moment a House Committee Member reports an issue to when it's resolved?
- Where in that process do things, most often go wrong or get delayed?
- How do you currently track which complaints are still open versus resolved?
- How do you decide which complaints get attended to first?
- What happens today if a maintenance job isn't completed on time is there a formal escalation step?
- How do you currently know whether a "completed" job solved the resident's problem?
- What's the biggest frustration you personally experience with the current process?

2. Meeting Notes:

Meeting with Estate and Infrastructure Unit CUT

Date: 19 August 2026

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Stakeholder: Mr Khumalo – Estate and Infrastructure Unit

Purpose: To understand the department's current system, challenges, and requirements for the proposed project.

Discussion

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Requirements Discussed

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- Keep proper records of reported issues.
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- Allow staff to update the status of a request, whether it is pending, in progress, or completed.
- Make it easier to see which issues still need attention.
- Keep information organised so that it can be accessed when needed.
- Improve communication between the people reporting problems and the Estate and Infrastructure Unit.

- Provide a simple and user-friendly interface.
- Protect the information stored in the system and ensure that only authorised people can access certain information.

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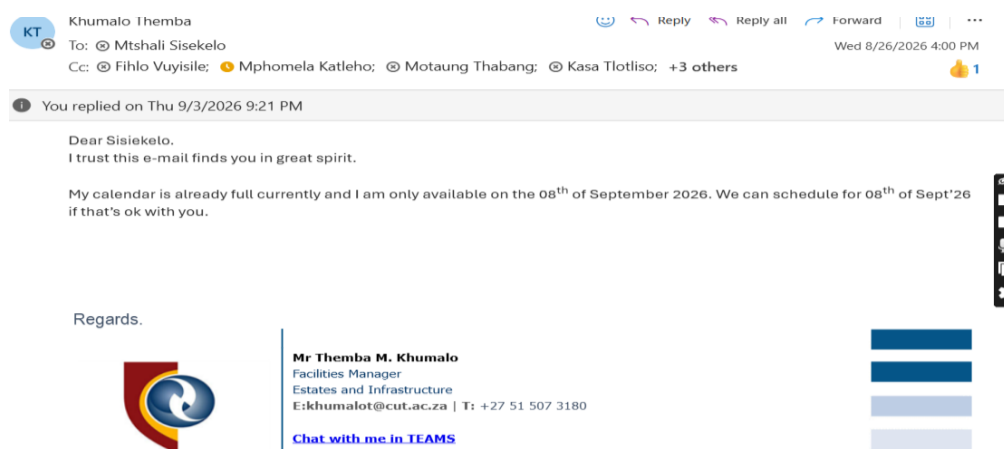
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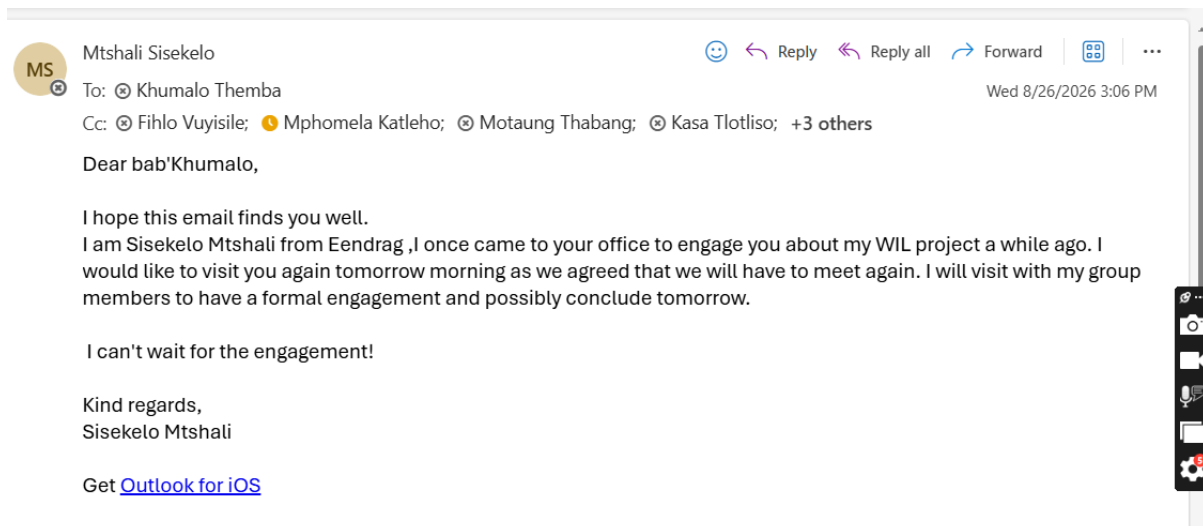
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Conclusion

The meeting was useful because Mr Khumalo gave us a better understanding of the Estate and Infrastructure Unit's current system, their challenges, and the requirements for the proposed system.

3. Email communication:





Conclusion

This SRS establishes a realistic, integrated prototype that directly addresses the identified CUT maintenance complaint-management problem. The core improvement is the movement from a fragmented, manual reporting process to a controlled digital lifecycle in which complaints can be reported, prioritised, assigned, monitored, escalated, completed, verified, closed or reopened, and retained as historical information.

The highest rated project risks are insufficient time, integration failure and incorrect deadline have directly shaped the refined scope.

Supabase Project

